Bees 2: Varroa mites and deformed wing virus

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```
ENV["GKSwstype"] = "100"
using MatrixCommunityModels
using Plots
```

The previous vignette built a honey bee colony. This one adds the mite *Varroa destructor* as a second species, and then deformed wing virus (DWV) as a second *stratum* of both species.

Mites are a two-stage net

A mite is either on an adult bee (`phoretic`) or in a capped brood cell, where it stays for the 12 days until the bee emerges and then leaves with its daughters.

```
q = VARROA_PARAMETERS
mite_life = ValuedProjectionNet([:phoretic, :cell],
    :waiting   => [(:phoretic => :phoretic) => 1.0],
    :invading  => [(:phoretic => :cell) => 1.0],
    :leaving   => [(:cell => :phoretic) => 1 - q.emergence_loss],
    :daughters => [(:cell => :phoretic) => q.offspring * (1 -
     ↪  q.emergence_loss)])
stage_names(expand_stage(mite_life, :cell, 12))
```

```
13-element Vector{Symbol}:
 :phoretic
 :cell1
 :cell2
 :cell3
 :cell4
 :cell5
 :cell6
 :cell7
 :cell8
 :cell9
 :cell10
 :cell11
 :cell12
```

`varroa()` is this net with three responses. The important one is `invading`: the fraction of phoretic mites entering cells each day depends on how many cells are being capped and how many adult bees there are, `1 - exp(-0.56 × cells capped / grams of adult bees)` (Boot et al. 1995; Calis et al. 1999).

Bees and mites meet through pools

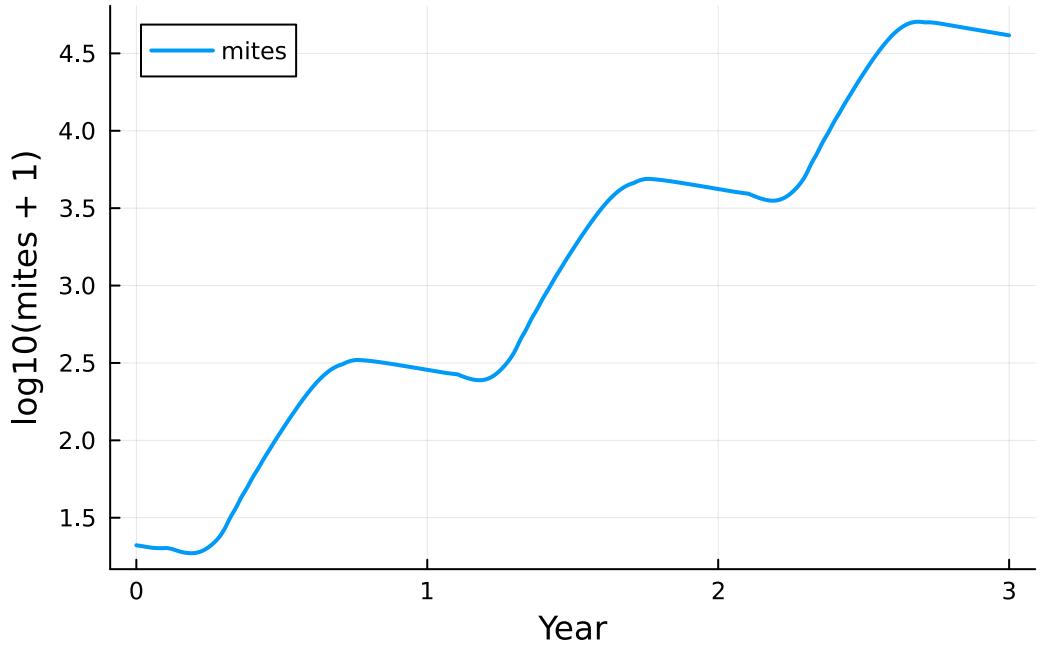
The mite never refers to the bee, or the bee to the mite. The bee reports three numbers as pools, and the mite reads them.

```
c = community(honeybee(), varroa())
pool_names(c)
```

```
4-element Vector{Symbol}:
 :adult_bees
 :capping
 :emerging
 :phoretic
```

`capping` is the number of larvae about to be capped, `adult_bees` the number of adults, `emerging` the number of pupae about to emerge, and `phoretic` the number of mites on adults.

```
sol = project(c, honeybee_state(c; mites = 20), 1:(365 * 3))
years = sol.t ./ 365
mites = abundance(sol, :varroa)
adults(s) = stage_total(s, :honeybee, [:hive, :forager])
plot(years, log10.(mites .+ 1); label = "mites", xlabel = "Year", ylabel =
 ↪  "log10(mites + 1)", lw = 2)
```




```
(growth_per_day = round(log(mites[211] / mites[151]) / 60; digits = 4),
 ↪  fold_in_first_year = round(mites[366] / 20; digits = 1))
```

```
(growth_per_day = 0.0185, fold_in_first_year = 14.2)
```

Martin (1998) reports 0.021 per day while there is brood and a 12-fold increase a year. Nothing here was tuned to that: the daughters per mite, the mites lost at emergence and the mortality are Martin's values. Drone brood, which mites prefer and in which they produce more daughters, is not modelled.

In this model mites alone do not harm the bees, following BEEHAVE, in which all damage comes through viruses:

```
alone = project(honeybee(), honeybee_state(honeybee()), 1:(365 * 3))
maximum(abs.(adults(sol) .- adults(alone)))
```

```
0.0
```

The virus is a stratum

DWV divides bees and mites into those without and with the virus. That is a stratification of each life cycle, not a new model: every stage and transition is replicated, and a *coupling* says which transitions can cross from one stratum to the other.

```
healthy_only = honeybee()
with_virus = honeybee(virus = true)
length(stage_names(healthy_only, :honeybee)), length(stage_names(with_virus,
 ↪  :honeybee))
```

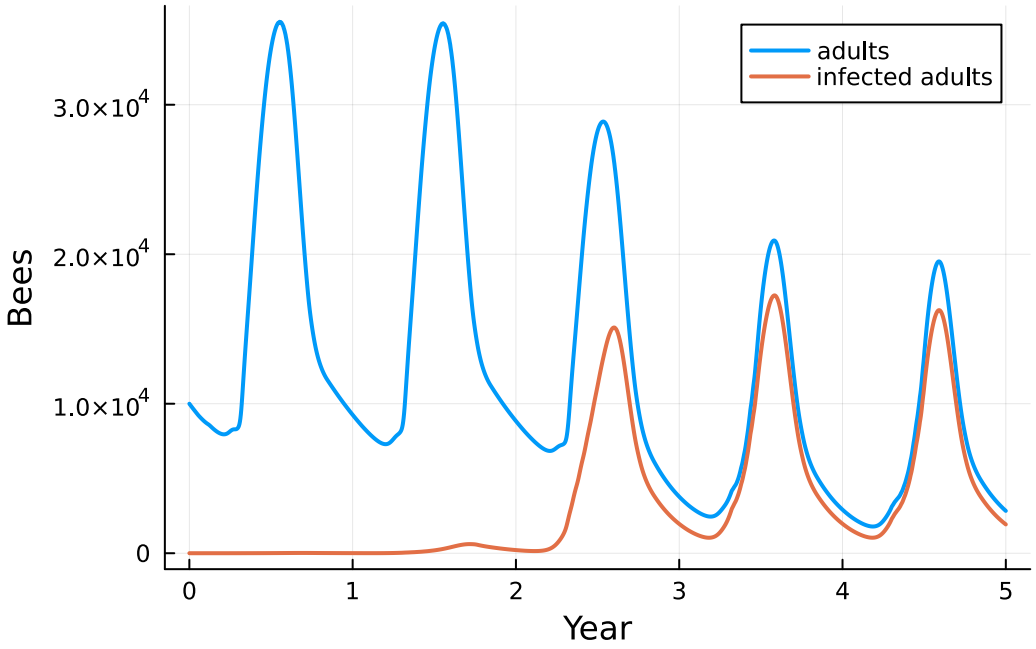
```
(24, 46)
```

```
filter(t -> occursin("_to_", String(t)),
 ↪  transition_names(system(with_virus)[:honeybee]))
```

```
4-element Vector{Symbol}:
 :capping_healthy_to_infected
 :foraging_healthy_to_infected
 :hive_life_healthy_to_infected
 :recruiting_healthy_to_infected
```

A larva crosses at capping, if a mite carrying the virus enters its cell; adults could also cross by mouth (used in the last vignette). The responses that differ for infected bees are attached to the transitions of the infected stratum: a fifth of infected pupae die or emerge deformed, and infected adults die three times as fast in the hive. Mites are stratified the same way (`clean`, `viral`): a clean mite becomes viral by sharing a cell with an infected pupa or by taking an infected adult as its host, and stays viral. A mite takes a new host when it leaves a cell, or when its host dies or becomes a forager; the chance that the new host is infected is the fraction of in-hive bees that are (the rule used by BEEHAVE).

```
function colony(; treatment = day -> 0.0, years = 5)
    c = community(honeybee(virus = true), varroa(virus = true, treatment =
 ↪  treatment))
    project(c, honeybee_state(c; mites = 20, viral = 0.1), 1:(365 * years))
end
untreated = colony()
infected(s) = stage_total(s, :honeybee, [:hive_infected, :forager_infected])
plot(untreated.t ./ 365, [adults(untreated) infected(untreated)]; label =
 ↪  ["adults" "infected adults"],
     xlabel = "Year", ylabel = "Bees", lw = 2)
```



For two years little happens. In the third autumn, when mites are numerous and brood is scarce, most of the bees that will form the winter cluster are infected as pupae. They die early, and the colony goes into each following winter with a few thousand bees.

```
winters = 365 .* (1:5) .+ 1
[(year = y, adults = round(Int, adults(untreated)[w]), mites_per_100_bees =
 ↪  round(100 * stage_total(untreated, :varroa, :phoretic)[w] /
 ↪  adults(untreated)[w]; digits = 1),
  infected = round(infected(untreated)[w] / adults(untreated)[w]; digits =
 ↪  2)) for (y, w) in enumerate(winters)]
```

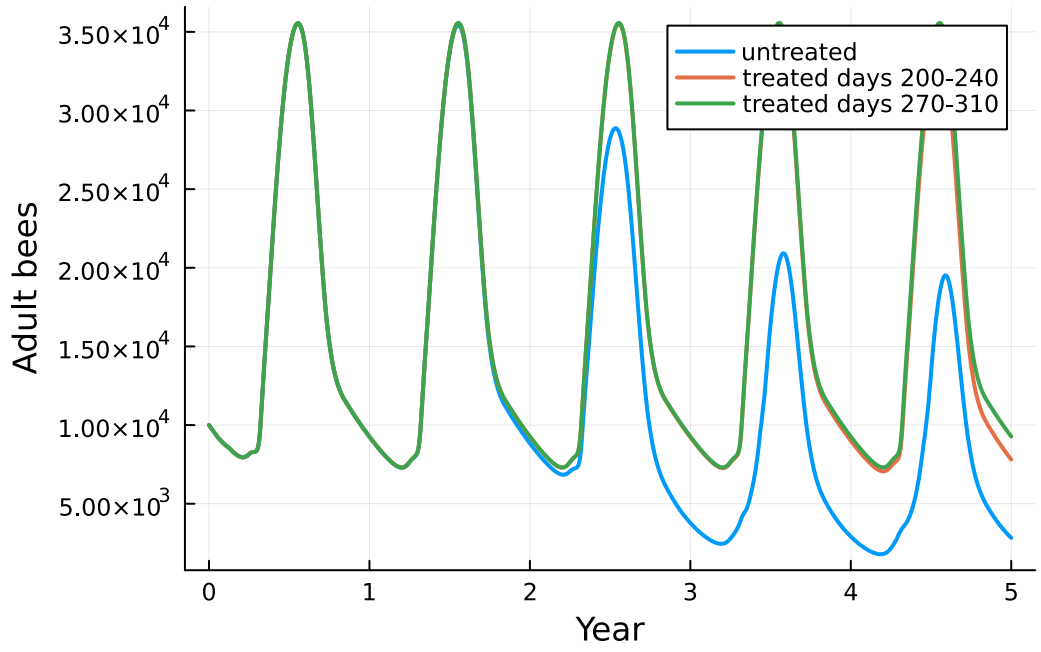
```
5-element Vector{@NamedTuple{year::Int64, adults::Int64,
 ↪  mites_per_100_bees::Float64, infected::Float64}}:
 (year = 1, adults = 9258, mites_per_100_bees = 3.1, infected = 0.0)
 (year = 2, adults = 8840, mites_per_100_bees = 47.8, infected = 0.02)
 (year = 3, adults = 3780, mites_per_100_bees = 1149.9, infected = 0.52)
 (year = 4, adults = 2913, mites_per_100_bees = 2402.1, infected = 0.68)
 (year = 5, adults = 2832, mites_per_100_bees = 2424.9, infected = 0.68)
```

A colony of fewer than about 4,000 bees is usually taken not to survive the winter (BEEHAVE; Kielmanowicz et al. 2015), so this colony is effectively lost in its third or fourth winter. Field studies put the autumn load of colonies that died at about 15 mites per 100 bees against 3 in survivors (Genersch et al. 2010).

Treatment, and its timing

A treatment is an extra mortality of phoretic mites over a window of days. Two windows of 40 days, each year:

```
late = colony(treatment = day -> 270 <= day < 310 ? 0.115 : 0.0)
early = colony(treatment = day -> 200 <= day < 240 ? 0.115 : 0.0)
plot(untreated.t ./ 365, [adults(untreated) adults(early) adults(late)];
     label = ["untreated" "treated days 200-240" "treated days 270-310"],
 ↪  xlabel = "Year", ylabel = "Adult bees", lw = 2)
```




```
[(treatment = name, winter_5_adults = round(Int, adults(s)[winters[5]]),
 ↪  mites = round(Int, abundance(s, :varroa)[winters[5]]))
 for (name, s) in (("none", untreated), ("days 200-240", early), ("days
 ↪  270-310", late))]
```

```
3-element Vector{@NamedTuple{treatment::String, winter_5_adults::Int64,
 ↪  mites::Int64}}:
 (treatment = "none", winter_5_adults = 2832, mites = 68705)
 (treatment = "days 200-240", winter_5_adults = 7810, mites = 2165)
 (treatment = "days 270-310", winter_5_adults = 9272, mites = 0)
```

Treating in October, when there is no brood for the mites to hide in, removes nearly all of them and keeps the colony healthy. The same treatment in late July and August kills only the mites on adults at the time; those in cells survive it, and the population recovers enough to cause damage

within a few years. (Field advice is to treat before the winter bees are reared; in this model the absence of brood matters more, which may reflect its lack of drone brood and of mites arriving from other colonies.)

Where the numbers come from

The mite and virus parameters are those of Martin (1998, 2001), checked against the parameter list and program code of BEEHAVE (Becher et al. 2014), which cite them: 1.01 daughters per mite in a worker cell, 30% of mites lost at emergence, mite mortality of 0.006 a day with brood and 0.002 without, transmission from mite to pupa of 0.89, death of a fifth of infected pupae, adult mortality of 0.012 a day after infection as a pupa, and a treatment killing 11.5% of phoretic mites a day for 40 days. They are listed in `VARROA_PARAMETERS` and `HONEYBEE_PARAMETERS`. Martin's papers themselves were not read.

Three things differ from those sources. Several mites sharing a cell leave fewer daughters each: Martin gives factors of 1, 0.91, 0.86 and 0.60 for one to four mites, and the smooth form used here gives 1, 0.91, 0.83 and 0.77. Mite mortality switches with the foraging season, not with the presence of brood. And viral mites do not infect adult bees, which BEEHAVE allows (without extra mortality for this virus).

References

Martin, S. J. (1998). *Ecological Modelling* 109, 267-281; (2001) *Journal of Applied Ecology* 38, 1082-1093. Becher, M. A. et al. (2014). *Journal of Applied Ecology* 51, 470-482. Genersch, E. et al. (2010). *Apidologie* 41, 332-352.